

Joann Jones

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EDUCATION

University of Chicago, Chicago, IL Aug 2024 - Present
Ph.D., Astronomy and Astrophysics (expected 2029-30)

Case Western Reserve University (CWRU), Cleveland, OH Aug 2020 - May 2024
B.A. Physics and Astronomy

Awards and Fellowships

LSST-DA Data Science Fellow (LSST-DA 2026-28)
Jump Trading Graduate Fellowship (Jump Trading 2025)
Kusch Prize for Top Senior Thesis in Physics (CWRU 2024)
Foldy Award for Top Outstanding Senior in Physics (CWRU 2024)
Nassau Prize for Outstanding Senior in Astronomy (CWRU 2024)
CWRU Board of Trustees Research Award Winner (CWRU 2023)
Wyant Award for Research Accomplishments Outside Senior Project (CWRU 2024)
Garwin Award for Service and Scholarship in Physics (CWRU 2024)
Eva L. Pancoast Memorial Fellowship for Research Travel (CWRU 2024)
Dean's High Honor List (CWRU 2020-24)

RESEARCH EXPERIENCE

Detectability of Thawing Dark Energy in LSST Supernova Data, UChicago Sept 2024 - Present
Advisor: Dr. Joshua Frieman

We investigate the viability of constraining a class of evolving and spatially varying dark energy models using forecasted Vera C. Rubin Large Synoptic Survey Telescope data.

Growth of Structure vs Geometry: Examination of Tensions Between the CMB and DESY6, UChicago Apr 2025 - Present
Advisor: Dr. Scott Dodelson

In attempt to understand current tensions between late and early time cosmological data, we analyze DESY6 3x2 data while setting a tight CMB prior on all cosmological parameters, except for σ_8 . This provides an interesting test of whether the tension is sourced by an issue with our models of structure growth or in the geometry of our measurements.

Uncorrelated Anomalies in the CMB, CWRU Mar 2023 - Present
Advisors: Dr. Glenn Starkman and Dr. Craig Copi

Using Python, simulated the Cosmic Microwave Background (CMB) to perform a statistical analysis of the "large angle anomalies." Found a p-value of Λ CDM reproducing the statistics in our CMB of 3×10^{-8} , over a 5σ deviation from the model. The results of this project have been posted to arxiv and submitted to Physical Review Letters.

Macroscopic Dark Matter Search, CWRU Jan 2022 - Aug 2023
Advisor: Dr. Glenn Starkman

Helped create a Citizens Science project to connect the search for dark matter with scientific outreach. Investigated machine learning techniques for image classification and worked with 3-D printing a custom frame.

Machine Learning to Detect Relativistic Electron Precipitation Bands, CU Boulder, Laboratory for Atmospheric and Space Physics May 2022 - Aug 2022
Advisors: Dr. Sergio Vidal-Luengo and Dr. Lauren Blum

Using Python, developed a machine learning algorithm to filter years of electron count data and automatically catalog relativistic electron precipitation band events. Utilized an unsupervised competitive neural network known as a self-organizing map.

PUBLICATIONS AND PRESENTATIONS

J. Jones, C. J. Copi, G. D. Starkman, Y. Akrami, “The Universe is not statistically isotropic”
arxiv:2310.12859 (submitted to Physical Review Letters) Oct 2023

Paper

Jones, J., Frieman, J., “Constraining Evolving and Spatially Varying Dark Energy” (University of Arizona Multiprobe Cosmology Winter School) Jan 2026

Poster

Jones, J., Frieman, J., “Constraining Evolving and Spatially Varying Dark Energy” (Carnegie Melon COSMO 2025) Oct 2025

Poster

Jones, J., Vidal-Luengo, S., Blum, L., “Using Machine Learning to Understand Relativistic Electron Precipitation,” (Brown University CUWiP) Jan 2023

Presentation

Jones, J., Vidal-Luengo, S., Blum, L., “Detecting Precipitation Bands Observed by SAMPEX using Power Spectral Density and a Self-Organizing-Map,” (AAS 241st Meeting 2023) Jan 2023

Poster

OUTREACH AND TEACHING

Community Engagement Group, UChicago Sept 2025 - Present

Active member of the CEG at UChicago. Led an effort to raise over \$3000 for food banks in the surrounding community.

NASA Partner Eclipse Ambassador, ASP Oct 2022 - April 2024

Funded by NASA and selected by the Astronomical Society of the Pacific, coordinated outreach events with local under-served communities with the goal to educate and provide resources for the upcoming total eclipse.

Stars Teaching Assistant, UChicago Sept 2024 - Dec 2024

Teaching Assistant for introductory course on the formation and evolution of stars. Led two weekly lab sections, review sessions, held weekly office hours, and graded assignments and lab reports.

Big Bang Teaching Assistant, UChicago Mar 2025 - Jun 2025

Teaching Assistant for introductory course on cosmology and the Big Bang Theory. Leading two weekly lab sections, holding weekly office hours, giving monthly lectures, and grading assignments and lab reports.

Astronomy Educator, Space Physics Jul 2022 - Aug 2022

Taught a summer camp course on astronomy. Prepared and taught classes on topics such as dark matter, planets, stars, black holes, and galaxies.

SKILLS

Computer Python, C++, Matlab, Mathematica, L^AT_EX

Languages English, German

Extras Forklift Certified :) (Narrow Isle Order Picker)